

Question	Answer	Marks
4(a)	pulses (of ultrasound from generator)	B1
	reflected at boundaries (between media)	B1
	time delay (between transmission and receipt) gives information about depth	B1
	intensity of reflected pulse gives information about nature (of tissues)/type (of tissues)/boundary	B1
	Any two from: • (reflected pulses) detected by the (ultrasound) generator • gel used to minimise reflection at skin/maximise transmission into skin • degree of reflection depends upon impedances of two media (at boundary)	B2
4(b)(i)	product of density and speed	M1
	speed of ultrasound in medium	A1
4(b)(ii)	Z_1 about equal to Z_2 results in negligible/no reflection	B1
	$Z_1 \gg Z_2$ (or $Z_1 \ll Z_2$) results in mostly reflection	B1

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5(c)	Any two reasonable suggestions e.g.:	22